Lesson

4 - 1

Types of Angles and Relations between Angles

■ Learning Outcomes

- Learn the concept of angles and their units of measurement.
- Identify and differentiate between various types of angles.
- Learn the adjacent angles.
- Learn the complementary angles.
- Learn the supplementary angles.
- Learn the vertically opposite angles.
- Learn about the accumulative angles at a point.
- Identify angles pairs in geometric shapes and the relationships between each pair.
- Apply the relationships between angles pairs to solve problems.

■ Vocabulary

- Straight Angle
- Reflex Angle
- Adjacent Angles
- Complementary Angles
- Supplementary Angles
- Vertically Opposite Angles
- Accumulative Angles at a Point

Get Ready!

Egypt has significant potential for generating clean, renewable energy from wind. The wind farm in Gabal El Zeit, south of Ras Gharib city, is one of the largest power plants to generate electricity in the world.

The opposite figure shows a wind turbine used to generate electricity. Can you determine the value of x?

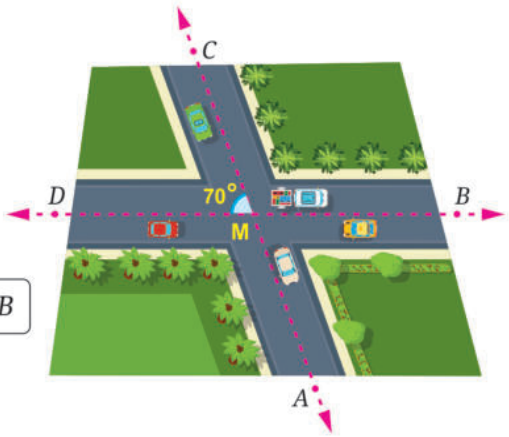


In this lesson, you will learn about the types of angles and the relationships between them, which will enable you to solve such real-life problems.

Think & Discuss!

The opposite figure represents two intersecting roads. If the measure of $\angle CMD$ is 70°, can you determine the measures of the following three angles?

$\angle AMD$, $\angle AMB$ and $\angle CMB$



Learn!

Concept of Angle: an angle is the union of two rays with the same initial point.

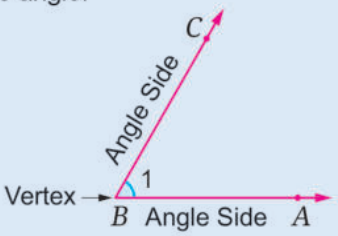
- The initial point of the two rays is called the vertex of the angle.
- Each one of the two rays is called side of the angle.

Example :

In the opposite figure:

$\overrightarrow{BA} \cup \overrightarrow{BC} = \angle ABC$

$\angle ABC$ can also be referred to as $\angle CBA$, $\angle B$ or $\angle 1$



Units of Measurement for Angles:

Units of measurement for angles are degrees, minutes, and seconds, such that:

- A degree is equal to 60 minutes (1° = 60').
- A minute is equal to 60 seconds (1' = 60").